

Black Box Methodology for Technical Compliance

with the EU Artificial Intelligence Act

Regulation (EU) 2024/1689 — Articles 5, 9-15, 50, 53, 55

Ahmad Qasim Mohammad Hassan

ORCID: 0009-0001-4360-0802

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Abstract

The EU Artificial Intelligence Act (Regulation EU 2024/1689) entered into force in February 2025 with obligations applying progressively. However, a fundamental gap persists between the binding legal text and actual technical verification: regulatory authorities currently rely on self-assessment documentation provided by companies without any independent technical tool to verify compliance. At the same time, direct regulatory access to corporate infrastructure is constrained by trade secret and data protection obligations.

This paper presents a technical compliance methodology that addresses this fundamental tension. The proposed approach requires companies to establish an internal compliance mechanism — a monitoring layer embedded within their existing infrastructure — that continuously records every violation with a tamper-protected cryptographic signature, and automatically sends a minimal alert (violation code, timestamp, record identifier) to regulatory authorities upon detection of any breach, without disclosing trade secrets or operational data.

The methodology covers all AI systems without restriction to any specific type, and translates ten articles of the EU AI Act into technically verifiable requirements grounded in peer-reviewed literature from ACM, IEEE, ISO, and NIST. The four-layer illustrative architecture is presented as a conceptual example requiring expert verification, not a final implementation standard.

1. Introduction

1.1 The Problem

The EU Artificial Intelligence Act is the world's first comprehensive legal framework for AI regulation. Its 113 articles address legislative, institutional, and administrative governance across all categories of AI systems. However, the majority of these articles impose obligations on Member States and the European Commission, not on the technical systems of companies.

Articles that directly address technical systems fall into two categories:

- Absolute prohibitions with no exceptions: Article 5 (eight prohibited practices)
- Mandatory technical requirements for high-risk systems: Articles 9 to 15, Article 50, and Articles 53 and 55

The fundamental problem is that compliance with these articles is currently verified through self-assessment reports and paper documentation. Regulatory authorities possess no independent technical tool to prove that an AI system actually behaves in accordance with these provisions. Direct access to corporate infrastructure is subject to legal and commercial constraints.

1.2 The Research Gap

No published methodology currently exists that translates the specific technical provisions of the EU AI Act — article by article, paragraph by paragraph — into automatically verifiable requirements embedded in corporate infrastructure. Existing compliance frameworks address either general legal interpretation or broad AI safety principles, but not the specific technical implementation of individual legal obligations.

1.3 Proposed Solution

This paper proposes a methodology whereby each company establishes an internal compliance layer — a structured monitoring mechanism embedded in its existing infrastructure — such that it:

- Continuously monitors system behavior according to rules defined for each legal article
- Records every violation in a tamper-protected cryptographic log that cannot be modified without detection
- Automatically sends a minimal alert — containing only the violation code, timestamp, and record identifier — to the regulatory authority upon detection of any breach
- Maintains full trade secrecy: regulatory authorities receive only the minimal alert, and may request the specific record under legal authority

1.4 Scope and Limitations

This methodology applies to all AI systems as defined in Article 3(1) of the Regulation, without restriction to any specific type. The illustrative four-layer architecture in Section 4 uses examples from large language model contexts for explanatory purposes; however, the underlying principles apply equally to computer vision systems, decision support systems, biometric systems, and recommendation engines.

The four-layer illustrative architecture is a conceptual example to demonstrate technical feasibility. It is not a final implementation specification, and requires review and verification by qualified software engineers, security specialists, and legal experts before any actual deployment. Specific numerical thresholds, catalogue

contents, and exception definitions are matters to be determined by competent authorities and domain experts.

Detailed translation of each article is available in supplementary documents provided upon request.

2. Methodology — Article Selection Criteria and Translation Approach

2.1 Article Selection Criteria

Of the 113 articles comprising the EU AI Act, this paper addresses only those that impose direct technical obligations on AI systems operating within corporate infrastructure. Articles relating to Member State obligations, institutional structures, financial penalties, and regulatory sandboxes are excluded.

Category	Articles	Nature of Obligation
Absolute prohibition	Article 5	8 prohibited practices for all systems without exception
High-risk requirements	Articles 9-15	Risk management, data governance, documentation, records, transparency, oversight, accuracy
User transparency	Article 50	Interaction notice, digital watermarking, analysis notice
GPAI obligations	Article 53	Documentation, copyright, information for downstream providers
Systemic risks	Article 55	Model evaluation, adversarial testing, reporting, energy efficiency

2.2 Translation Approach

This paper applies a systematic three-step translation process to each selected article:

Step	Description
1 — Legal Text	The legal provision is identified and formulated precisely as stated in the official text of Regulation (EU) 2024/1689
2 — Technical Requirement	The provision is translated into a technically verifiable requirement, expressed in terms of system behaviors, data fields, and observable conditions
3 — Scientific Basis	Each technical requirement is grounded in peer-reviewed literature from ACM, IEEE, ISO, or NIST to establish the scientific basis for the detection or enforcement mechanism

This translation methodology is the original contribution of this paper. The legal text is in the public domain. The peer-reviewed references are published works. The translation — the systematic mapping, article by article and paragraph by paragraph, of legal provisions to specific technically verifiable requirements grounded in published science — is the author's original work.

3. Technical Translation Article by Article

3.1 Article 5 — Prohibited Practices

Article 5 of Chapter II is the absolute red line of the Act. It prohibits eight types of practices for all AI systems without exception. Non-compliance admits no interpretation or mitigation.

Paragraph (a) — Subliminal and Manipulative Techniques

Legal Provision: Prohibition of AI systems that employ subliminal techniques beyond a person's consciousness, or exploitative or deceptive techniques, with the objective or effect of materially distorting behavior.

Technical Requirement: Every output (visual, auditory, textual, decisional) must be observable and measurable against human perception thresholds. Outputs operating below conscious awareness thresholds, or deceptive UI patterns that systematically constrain choice, trigger immediate blocking and logging.

Element	Value
Trigger	Any system output (visual, auditory, textual)
Predicate	Is FPS outside human perception range? Or Hz outside auditory range? Or did options drop to one without justification?
Enforcement	BLOCK + LOG type (a) in IAA + immediate alert

Scientific Basis: Gray et al. (ACM CHI 2024) establishes objective structural metrics for deceptive patterns. Mansur et al. (IEEE/ACM ICSE 2023) presents a four-stage automated detection pipeline. Kapoor et al. (Nature Communications 2022) defines the scientific threshold of conscious awareness in measurable neural terms.

Paragraph (b) — Exploitation of Vulnerability of Protected Groups

Legal Provision: Prohibition of AI systems that exploit the vulnerabilities of individuals due to age, physical or mental disability, or socioeconomic status, in a manner that materially distorts their behavior and causes harm.

Technical Requirement: Indicators of protected characteristics in input data must be monitored, and statistically significant behavioral changes in system outputs following such detection must be tracked. Discriminatory pricing or option narrowing linked to protected group membership must be blocked and logged.

Element	Value
Trigger	Request bearing a protected characteristic flag (age, disability, social status)
Predicate	Did system behavior change in a statistically measurable way after characteristic detection?
Enforcement	BLOCK + MODIFY + LOG type (b)

Scientific Basis: Loefflad & Grossklags (ACM FAccT 2024) provide specific metrics for behavioral impact on vulnerable groups. ISO 22458:2022 establishes the international standard for defining and protecting vulnerable consumers.

Paragraph (c) — Prohibited Social Scoring

Technical Requirement: The original context of data collection must be tracked, and cross-context data transfers must be detected. Any use of a temporal behavioral profile in a context that differs substantially from its original context must be blocked and logged.

Scientific Basis: Loefflad, Chen & Grossklags (AAAI/ACM AIES 2024) define social scoring and its measurable behavioral effects.

Paragraph (d) — Criminal Risk Assessment Without Objective Basis

Technical Requirement: Every criminal risk assessment request must include a mandatory structured field containing documented objective facts. Absence of this field, or requests relying exclusively on personal demographic characteristics, triggers immediate blocking.

Scientific Basis: Angwin et al. (ProPublica 2016) provides the most-cited evidence of actual harm from biased criminal assessment tools. Dressel & Farid (Science Advances 2018) demonstrate that crime prediction models do not outperform random human judgment.

Paragraphs (e) through (h)

The same translation methodology applies to the remaining prohibited practices:

- (e) Untargeted facial image collection: Block import from the internet or CCTV feeds without an individual consent record
- (f) Emotion inference in work and education: BLOCK when an emotion model is activated in an institutional context outside documented medical or security purposes
- (g) Biometric classification of sensitive categories: Immediate BLOCK when any of {race, religion, political_opinion, sexual_orientation, trade_union} appears in classification output
- (h) Real-time remote biometric identification in public spaces: BLOCK until a valid judicial authorization number is presented and documented by the competent judicial authority

3.2 Articles 9-15 — Requirements for High-Risk Systems

Article 9 — Risk Management System

Technical Requirement: A risk management system is not a document written once. It is a live operational structure running from design day to shutdown day with automated auditable records. It includes:

- A structured risk register with automated updates linked to the behavioral monitor
- A periodic monitor running at defined intervals, automatically adding results to the register
- Threat Modeling prior to launch in three categories: confirmed, probable, and speculative risks
- Testing against pre-specified criteria before launch — no ad hoc testing

Scientific Basis: NIST AI RMF 1.0 (2023) defines the risk management cycle in four functions: Govern, Map, Measure, and Manage. ISO/IEC 23894:2023 is the first international standard specifically addressing AI risk management.

Article 10 — Data Governance

Technical Requirement: Every dataset used in training or testing the system must carry a standardized document (Datasheet) specifying source, collection method, date, language, size, known limitations, and licensing method. The statistical representation of every protected category must be measured.

Scientific Basis: Gebru et al. (ACM CACM 2021) — Datasheets for Datasets. Torralba & Efros (IEEE CVPR 2011) demonstrate that most large datasets carry structural bias resulting from collection methodology.

Article 11 — Technical Documentation

Technical Requirement: Technical documentation is a package of seven documents prepared before launch and updated automatically:

- System description document (what it does, what it does not do)
- Model Card — Mitchell et al. 2019
- Datasheets for Datasets
- Risk management system results (Article 9)
- Test Results against pre-specified criteria
- Description of human oversight procedures (Article 14)
- Regulatory authority contact information

The package is stored in a format deliverable within 48 hours upon any regulatory request.

Article 12 — Record-Keeping

Technical Requirement: An automated operational log recorded immediately upon every event without exception. Every record carries a cryptographic hash of its

content including the previous record's hash, forming a Hash Chain — any modification to any record changes the hash and immediately reveals tampering.

Scientific Basis: Merkle (CRYPTO 1987) — the mathematical basis for hash chains (Merkle trees). NIST SP 800-92 specifies the minimum content of logs. ISO/IEC 27001:2022 — A.8.15.

Article 13 — Transparency

Technical Requirement: Every output is accompanied by a confidence score and simplified explanation of the reason. Low-confidence outputs are automatically escalated for human review. Every decision affecting a person is accompanied by a notice containing: This decision was made with AI assistance — you have the right to request an explanation and human review.

Scientific Basis: Ribeiro et al. LIME (KDD 2016). Lundberg & Lee SHAP (NeurIPS 2017). Wachter, Mittelstadt & Russell (Harvard JOLT 2017) — the individual's right to know the automated decision.

Article 14 — Human Oversight

Technical Requirement: Three-tier oversight classification:

Risk Level	Decision Mechanism	System Role
Low	System decides and logs	For subsequent review
Medium	System proposes, human approves	Decision support
High	Human decides only	Information display only

Emergency stop button available at all times without technical barriers. Live monitoring dashboard displaying system status, confidence indicators, and anomalies.

Scientific Basis: Parasuraman, Sheridan & Wickens (IEEE SMC 2000) — ten-level automation model. Shneiderman (Oxford University Press 2022) — Human-Centered AI.

Article 15 — Accuracy, Robustness, and Cybersecurity

Technical Requirement (Accuracy): Pre-specified accuracy metric (Accuracy, F1, AUC), measured on an independent test set representing the actual deployment environment. Continuous monitor re-measuring periodically and alerting upon decline.

Technical Requirement (Robustness): Testing against corrupted and contradictory inputs. Adversarial testing (Red Teaming) by an independent team before each launch. Every discovered vulnerability closed before launch.

Technical Requirement (Cybersecurity): Application of ISO/IEC 27001 to every component. Penetration testing every six months. Any high-risk vulnerability addressed within 72 hours.

Scientific Basis: Goodfellow et al. (ICLR 2015) — the founding paper on adversarial examples. Carlini & Wagner (IEEE S&P 2017). ISO/IEC 27001:2022. NIST CSF 2.0.

3.3 Article 50 — Transparency Toward the User

- Paragraph 1: Clear notice at the start of every interactive session that the user is interacting with an AI system — non-concealable, logged in IAA
- Paragraph 2: Every generated output (image, audio, long text) carries a digital watermark pursuant to the C2PA v2.1 standard
- Paragraph 3: Prior notice to the individual before any emotional or biometric analysis, with consent or refusal recorded — refusal triggers immediate BLOCK

Scientific Basis: Amershi et al. (ACM CHI 2019) — Guideline 12 on system identity disclosure. Fernandez et al. (IEEE/CVF ICCV 2023) — The Stable Signature. Kirchenbauer et al. (ICML 2023) — watermarking of generated text.

3.4 Article 53 — Obligations of GPAI Providers

General-Purpose AI (GPAI) models are models trained on large quantities of data capable of performing multiple tasks. Their obligations include:

- Comprehensive technical documentation: model architecture, training data, capabilities and limitations, benchmark evaluation results
- Delivery package for each downstream provider in the chain including permitted use scope, known limitations, and incident contact information
- Copyright compliance policy with opt-out mechanism for rights holders
- Public summary of training data published openly — source categories, time period, languages, what was explicitly excluded

Scientific Basis: Liang et al. HELM (Stanford CRFM 2022). Bommasani et al. Foundation Models (arXiv 2021). Henderson et al. — Foundation Models and Fair Use (arXiv 2023).

3.5 Article 55 — GPAI with Systemic Risks

Models exceeding the specified computation threshold (10^{25} floating point operations) or classified by the AI Office based on their capabilities are subject to additional obligations:

- Evaluation using HELM and BIG-bench standards before each launch with comparison to previous version results
- Independent adversarial testing (Red Teaming) documented — team attempts to produce harmful content, exploit outputs, defeat safety constraints

- Reporting of any serious incident to the regulatory authority within 72 hours of discovery
- Periodic reports on energy consumption and carbon footprint for each phase (training, fine-tuning, inference)

Scientific Basis: Perez et al. Anthropic (arXiv 2022) — Red Teaming with language models. Ganguli et al. Anthropic (arXiv 2022). Zou et al. CMU/UW (arXiv 2023) — universal and transferable adversarial attacks.

4. Illustrative Architecture — Conceptual Example

Notice: This section presents a four-layer architecture as a conceptual illustration only. This architecture is not a final design, a recommended standard, or a verified implementation. Any actual deployment requires review by qualified software engineers, cybersecurity specialists, and legal experts.

The architecture derives from a single core requirement: the system must be able to monitor compliance with each legal article without needing to read conversation content or expose personal data.

4.1 Layer One — Application Interception Proxy (EIP)

A reverse proxy placed between the AI system and its users. Every system output passes through this layer before delivery. EIP intercepts outputs for inspection without reading conversational content — it analyzes structural and behavioral characteristics, not semantic meaning.

Reference: Rebedea et al. NeMo Guardrails (NVIDIA, ACM EMNLP 2023) — defines the pipeline architecture and programmable rails.

4.2 Layer Two — Rule Engine (RE)

A decision mechanism that converts each legal provision into an executable rule in three-part form: Trigger + Predicate → Enforcement (BLOCK / MODIFY / ESCALATE / LOG). Rules are stored in a configuration file separate from system code.

Reference: Wang, Poskitt & Sun — AgentSpec (IEEE/ACM ICSE 2026) — defines the Trigger-Predicate-Enforcement structure as an implementation framework.

4.3 Layer Three — Behavioral Monitor (BM)

A statistical monitoring component that tracks patterns of system behavior over time rather than evaluating individual requests in isolation. Maintains rolling statistical windows and detects anomalous patterns — progressive option narrowing, time pressure patterns, output distribution drift.

Reference: Gama et al. — Survey on Concept Drift Adaptation (ACM Computing Surveys 2014).

4.4 Layer Four — Immutable Audit and Alert (IAA)

A tamper-protected logging system coupled with an automated regulatory notification mechanism. Every event is recorded with a cryptographic hash. Every record includes the previous record's hash forming a Hash Chain — any modification is immediately detected through chain integrity verification.

Upon detection of a violation, IAA automatically sends an encrypted alert to the regulatory authority containing exactly three fields:

Field	Value	Purpose
violation_code	Violation type code (e.g., ART5_A)	Identify the violated article and paragraph
timestamp	Precise unmodifiable timestamp	Establish the time of the incident
record_id	Record identifier in the IAA database	Enables the regulatory authority to request the record under legal authority

Reference: Merkle (CRYPTO 1987) — mathematical basis for hash chains. ISO/IEC 27001:2022 — Annex A.5.24. NIST AI RMF 1.0 GOVERN 1.7.

4.5 Mechanism for Resolving the Fundamental Regulatory Tension

The minimal three-field alert resolves the fundamental tension in AI Act enforcement:

Property	Details
Regulatory assurance	Every violation is automatically reported. Competent authorities receive real-time notification of every compliance breach
Commercial protection	The alert contains no operational data, system architecture information, or user data
Legal enforceability	The cryptographic hash chain ensures records cannot be retroactively modified
Auditability	Competent authorities monitor compliance through violation notifications rather than continuous access to company systems

5. Scope of Determination

5.1 What This Paper Determines

- Mapping of specific legal provisions to technically verifiable requirements (article, paragraph, and sub-paragraph)
- The peer-reviewed scientific basis for each technical requirement
- The conceptual architecture demonstrating the technical feasibility of embedded continuous compliance monitoring

- Design of the three-field minimal alert mechanism as a solution to the tension between regulatory oversight and trade secrecy

5.2 What This Paper Does Not Determine

The following values are to be determined by competent authorities and domain experts:

- Specific numerical thresholds for any detection parameter (FPS limits, Hz ranges, statistical deviation limits)
- Precise contents of the Annex I catalogue (20 harmonized legislative instruments) and Annex III catalogue (8 sectoral categories)
- Precise legal definitions of exceptions under Article 6(3)
- Specific data fields constituting the objective factual basis for criminal risk assessments
- Registration protocol in the EU AI database under Article 71

6. Original Contribution

The original contribution of this paper is the structured methodology for translating EU AI Act provisions into technically verifiable compliance requirements embedded in infrastructure. Specifically:

- The translation methodology (legal text → technical requirement → peer-reviewed scientific basis) applied consistently across ten articles
- Identification of the tension between regulatory oversight and trade secrecy as the fundamental problem in Act enforcement, and the three-field minimal alert mechanism as a solution
- Conceptual proof that continuous automated tamper-protected compliance verification is technically feasible without requiring direct regulatory access

The text of the EU AI Act is in the public domain. The peer-reviewed references are published works. The standards (ISO and NIST) are public standards. The translation — the systematic mapping, article by article and paragraph by paragraph, of legal provisions to specific technically verifiable requirements with a scientific basis — is the original work of the author.

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